

HONDA

V45/65 Sabre & Magna

(VF700, 750 & 1100 V-Fours)

1982 to 1988 □ 699cc □ 748cc □ 1098cc

Owners Workshop Manual



820



Honda

V45/65 Sabre & Magna (VF700, 750 & 1100 V-Fours)

Owners Workshop Manual

by Penny Cox
and John H. Haynes

Member of the Guild of Motoring Writers

Models covered:

Honda V45 700/750 Sabre. US 1982 through 1985
Honda VF750S-C Sport. UK 1982 through 1984
Honda V45 700/750 Magna. US 1982 through 1988
Honda VF750C-H and C-J Custom. UK 1987 through 1988
Honda V65 1100 Sabre. US 1984 through 1985
Honda V65 1100 Magna. US 1983 through 1986



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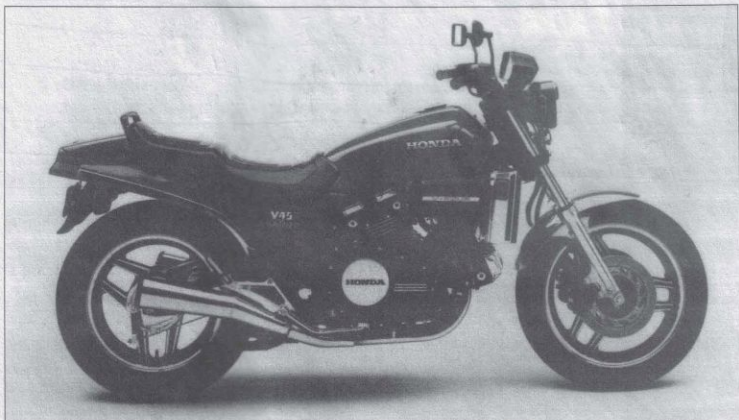
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The 1982 V45 Sabre (VF750S)



The 1984 V65 Sabre (VF1100S)



The 1982 V45 Magna (VF750C)



The 1984 V65 Magna (VF1100C)

About this manual

Its purpose

The purpose of this manual is to help you get the best value from your motorcycle. It can do so in several ways. It can help you decide what work must be done, even if you choose to have it done by a dealer service department or a repair shop; it provides information and procedures for routine maintenance and servicing; and it offers diagnostic and repair procedures to follow when trouble occurs.

We hope you use the manual to tackle the work yourself. For many simpler jobs, doing it yourself may be quicker than arranging an appointment to get the vehicle into a shop and making the trips to leave it and pick it up. More importantly, a lot of money can be saved by avoiding the expense the shop must pass on to you to cover its labor and overhead costs. An added benefit is the sense of satisfaction and accomplishment that you feel after doing the job yourself.

Using the manual

The manual is divided into Chapters. Each Chapter is divided into numbered Sections, which are headed in bold type between horizontal lines. Each Section consists of consecutively numbered paragraphs or steps.

At the beginning of each numbered Section you will be referred to any illustrations which apply to the procedures in that Section. The reference numbers used in illustration captions pinpoint the pertinent Section and the Step within that Section. That is, illustration 3.2 means the illustration refers to Section 3 and Step (or paragraph) 2 within that Section.

Procedures, once described in the text, are not normally repeated. When it's necessary to refer to another Chapter, the reference will be given as Chapter and Section number. Cross references given without use of the word 'Chapter' apply to Sections and/or paragraphs in the same Chapter. For example, 'see Section 8' means in the same Chapter.

References to the left or right side of the vehicle assume you are sitting on the seat, facing forward.

Motorcycle manufacturers continually make changes to specifications and recommendations, and these, when notified, are incorporated into our manuals at the earliest opportunity.

Even though we have prepared this manual with extreme care, neither the publisher nor the author can accept responsibility for any errors in, or omissions from, the information given.

NOTE

A **Note** provides information necessary to properly complete a procedure or information which will make the procedure easier to understand.

CAUTION

A **Caution** provides a special procedure or special steps which must be taken while completing the procedure where the Caution is found. Not heeding a Caution can result in damage to the assembly being worked on.

WARNING

A **Warning** provides a special procedure or special steps which must be taken while completing the procedure where the Warning is found. Not heeding a Warning can result in personal injury.

Introduction to the Honda V45/65 Sabre & Magna (VF700, 750 & 1100 V-Fours)

The first Honda V-Four engine, introduced in 1982 in a 750 cc (45 cu in) capacity, was widely regarded as a milestone of motorcycle engineering. Its 90° V configuration allowed for a lighter engine which was more compact, yet more powerful than any previous Honda 750 cc engine. Because of the inherent balancing characteristics of a V-engine, as well as the rubber mountings, it is also an unusually smooth and quiet engine compared with the more conventional in-line, air-cooled counterparts in production at the time.

The 750 cc engine was superseded in 1984 by a shorter-stroke 700 cc model, designed to fall below the heavyweight motorcycle import tariff imposed on machines imported into the US. The 700 cc engine continued for the remaining years of the Sabre, but a return was made to the 750 cc unit in 1988 for the last year of the Magna's production. All models imported into the UK were of 750 cc capacity.

An 1100 cc (65 cu in) engine was introduced in 1983 for the US market.

There are two distinctly different models, the Sabre (known as the Sport in the UK) and the Magna (known as the Custom in the UK). The Sabre differs from the Magna in having Honda's Pro-Link rear suspension and sophisticated electronic instrumentation. Conversely, Magna models are custom-styled, with high handlebars, teardrop tank and a stepped seat. The Magnas have conventional twin-shock rear suspension.

Both Sabre and Magna have appeared in 700, 750 and 1100 cc engine sizes during the model run, and have received a number of improvements and modifications. Owners are therefore advised to refer to the table under 'Identification numbers' to establish the exact model year of their machine before carrying out any of the procedures given in the main text of the manual.

Identification numbers

The frame serial number is stamped into the right side of the steering head and the VIN (Vehicle Identification Number) appears on the left side of the steering head; on 1987 and 1988 700/750 Magna models, it is attached to the right frame top tube under the fuel tank. The engine number is stamped into the right upper side of the crankcase, directly above the clutch unit.

A label attached to the right or left frame tube under the side cover, or attached to the rear fender top surface under the seat, gives the color code of the machine. The carburetor identification number appears on the carburetor body casting, just above the float chamber joint. Emission control information (US models only) is given on a label attached to the right lower frame tube on models through 1986, or to the right upper frame tube on later models.

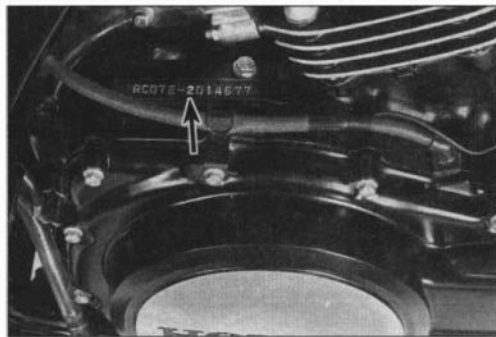
All serial numbers should be recorded and kept in a safe place so they can be furnished to law enforcement officials in the event of a theft.

The frame serial number, engine serial number, carburetor identification number and color code should also be kept in a handy place (such as with your driver's license) so they are always available when purchasing or ordering parts for your machine.

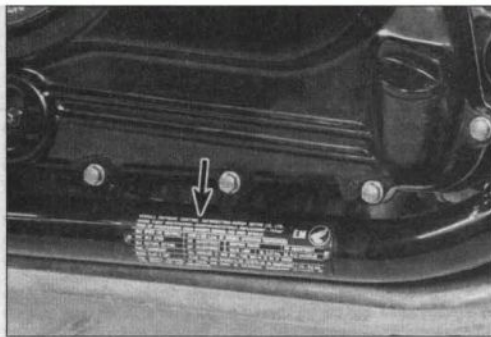
The procedures in this manual identify the bikes by production year. If this is not known, it can be determined from the engine and frame serial numbers as shown in the chart on the next page.



Frame number location on steering head right side



Engine number location on crankcase top surface



Emission Control label location (models through 1986)

US models (except California)

Engine number	Frame number
VF750S (1982)	RC07E-2000068 to 2015996
VF750S (1983)	RC07E-2100018 to 2115544
VF700S (1984)	RC22E-2000008 to 2005978
VF700S (1985)	RC22E-2100001 to 2103668
VF750C (1982)	RC07E-4000046 to 4029232
VF750C (1983)	RC07E-4100013 to 4124548
VF700C (1984)	RC21E-2000021 to 2016382
VF700C (1985)	RC21E-2100016 to 2110238
VF700C (1986)	RC21E-2200005 to 2209873
VF700C (1987)	RC21E-2300015 to 2304999
VF750C (1988)	RC07E-4600001 on
VF1100S (1984)	SC17E-2000001 to 2008727
VF1100S (1985)	SC17E-2100001 on
VF1100C (1983)	SC12E-2000039 to 2018597
VF1100C (1984)	SC12E-2100001 to 2116075
VF1100C (1985)	SC12E-2200001 to 2208055
VF1100C (1986)	SC12E-2300020 to 2304336
RC070-CM000036 to 012083	
RC070-DM100006 to 108632	
RC220-EM000007 to 005012	
RC220-FM100001 to 102800	
RC071-CM000033 to 027062	
RC071-DM100011 to 122819	
RC210-EM000002 to 014552	
RC210-FM100006 to 108900	
RC210-GM200003 to 208928	
RC210-HM300007 to 303508	
or HA305001 to 310415	
RC280-JA100001 on	
SC170-EM000001 to 010029	
SC170-FA100001 on	
SC120-DM000029 to 017677	
SC120-EA100001 to 117064	
SC120-FA200006 to 208465	
SC120-GM300101 to 304425	

US California models

VF750S (1982)	As above	As above
VF750S (1983)	As above	As above
VF700S (1984)	RC22E-2002575 to 2006097	RC221-EM000006 to 001081
VF700S (1985)	RC22E-2100001 to 2103285	RC221-FM100001 to 100863
VF750C (1982)	As above	As above
VF750C (1983)	As above	As above
VF700C (1984)	RC21E-2000019 to 2018662	RC211-EM000003 to 002300
VF700C (1985)	RC21E-2100022 to 2108674	RC211-FM100007 to 101325
VF700C (1986)	RC21E-2200011 to 2210193	RC211-GM200001 to 201250
VF700C (1987)	RC21E-2300020 to 2303889	RC211-HM300006 to 300514
		or HA305003 to 311015
VF750C (1988)	RC07E-4600001 on	RC281-JA100001 on
VF1100S (1984)	SC17E-2000001 to 2009087	SC170-EA100004 to 117067
VF1100S (1985)	SC17E-2100001 on	SC170-FA100001 on
VF1100C (1983)	As above	As above
VF1100C (1984)	SC12E-2102298 to 2114835	SC120-EA100004 to 117067
VF1100C (1985)	SC12E-2200001 to 2206580	SC120-FA200001 to 206845
VF1100C (1986)	SC12E-2300001 to 2302532	SC120-GM301306 to 302825

UK models

VF750S-C (1982-84)	RC07E-2007411 to 2020350	RC07-2000034 to 2008149
VF750C-H (1987)	RC07E-4501116 to 4501315	RC28-2000607 to 2000806
VF750C-J (1988)	RC07E-4604226 on	RC28-2100005 on

Note: Unless specifically mentioned in this manual, the information given for the 1982 750 Sabre applies to the UK VF750S-C, and that for the 1987 and 1988 700/750 Magnas applies to the UK VF750C-H and C-J respectively.

Buying parts

Once you have found all the identification numbers, record them for reference when buying parts. Since the manufacturers change specifications, parts and vendors (companies that manufacture various components on the machine), providing the ID numbers is the only way to be reasonably sure that you are buying the correct parts.

Whenever possible, take the worn part to the dealer so direct comparison with the new component can be made. Along the trail from the manufacturer to the parts shelf, there are numerous places that the part can end up with the wrong number or be listed incorrectly.

The two places to purchase new parts for your motorcycle - the accessory store and the franchised dealer - differ in the type of parts they carry. While dealers can obtain virtually every part for your

motorcycle, the accessory dealer is usually limited to normal high wear items such as shock absorbers, tune-up parts, various engine gaskets, cables, chains, brake parts, etc. Rarely will an accessory outlet have major suspension components, cylinders, transmission gears, or cases.

Used parts can be obtained for roughly half the price of new ones, but you can't always be sure of what you're getting. Once again, take your worn part to the wrecking yard (breaker) for direct comparison.

Whether buying new, used or rebuilt parts, the best course is to deal directly with someone who specializes in parts for your particular make.

General specifications

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Wheelbase

1982 750 Sabre model.....	1562 mm (61.5 in)
1983 through 1985 700/750 Sabre models.....	1570 mm (61.8 in)
1982 through 1984 700/750 Magna models.....	1540 mm (60.6 in)
1985 and 1986 700 Magna models.....	1565 mm (61.6 in)
1987 and 1988 700/750 Magna models.....	1660 mm (65.4 in)
1100 Sabre models.....	1590 mm (62.6 in)
1100 Magna models.....	1595 mm (62.8 in)

Overall length

700/750 Sabre models.....	2245 mm (88.4 in)
1982 through 1984 700/750 Magna models.....	2190 mm (86.2 in)
1985 and 1986 700 Magna models.....	2220 mm (87.4 in)
1987 and 1988 700/750 Magna models.....	2360 mm (92.9 in)
1100 models.....	2280 mm (89.8 in)

Overall width

1982 through 1984 700/750 Sabre models.....	830 mm (32.7 in)
1985 700 Sabre model.....	800 mm (31.5 in)
1982 through 1984 700/750 Magna models.....	815 mm (32.1 in)
1985 and 1986 700 Magna models.....	850 mm (33.5 in)
1987 and 1988 700/750 Magna models.....	800 mm (31.5 in)
1100 Sabre models.....	790 mm (31.1 in)
1983 and 1986 1100 Magna models.....	810 mm (31.9 in)
1984 and 1985 1100 Magna models.....	825 mm (32.5 in)

Overall height

1982 and 1983 750 Sabre models.....	1165 mm (45.9 in)
1984 and 1985 700 Sabre models.....	1160 mm (45.7 in)
1982 through 1984 700/750 Magna models.....	1195 mm (47.0 in)
1985 and 1986 700 Magna models.....	1200 mm (47.2 in)
1987 and 1988 700/750 Magna models.....	1155 mm (43.9 in)
1100 Sabre models.....	1185 mm (46.7 in)
1983 and 1986 1100 Magna models.....	1210 mm (47.6 in)
1984 and 1985 1100 Magna models.....	1230 mm (48.4 in)

Seat height

1982 and 1983 750 Sabre models.....	780 mm (30.7 in)
1984 and 1985 700 Sabre models.....	790 mm (31.1 in)
1982 through 1984 700/750 Magna models.....	760 mm (29.9 in)
1985 and 1986 700 Magna models.....	740 mm (29.1 in)
1987 and 1988 700/750 Magna models.....	705 mm (27.8 in)
1100 Sabre models.....	820 mm (32.3 in)
1100 Magna models.....	800 mm (31.5 in)

Ground clearance

1982 and 1983 750 Sabre models.....	135mm (5.3 in)
1984 and 1985 700 Sabre models.....	145 mm (5.7 in)
1982 through 1984 700/750 Magna models.....	165 mm (6.5 in)
1985 and 1986 700 Magna models.....	150 mm (5.9 in)
1987 and 1988 700/750 Magna models.....	152 mm (6.0 in)
1100 Sabre models.....	145 mm (5.7 in)
1100 Magna models.....	155 mm (6.1 in)

Weight (with oil and full fuel tank)

700/750 models.....	Approx 243 kg (535 lb)
1100 Sabre models.....	268 kg (591 lb)
1100 Magna models.....	265 kg (584 lb)

Maintenance techniques, tools and working facilities

Basic maintenance techniques

There are a number of techniques involved in maintenance and repair that will be referred to throughout this manual. Application of these techniques will enable the amateur mechanic to be more efficient, better organized and capable of performing the various tasks properly, which will ensure that the repair job is thorough and complete.

Fastening systems

Fasteners, basically, are nuts, bolts and screws used to hold two or more parts together. There are a few things to keep in mind when working with fasteners. Almost all of them use a locking device of some type (either a lock washer, locknut, locking tab or thread adhesive). All threaded fasteners should be clean, straight, have undamaged threads and undamaged corners on the hex head where the wrench fits. Develop the habit of replacing all damaged nuts and bolts with new ones.

Rusted nuts and bolts should be treated with a penetrating oil to ease removal and prevent breakage. Some mechanics use turpentine in a sprout type oil can, which works quite well. After applying the rust penetrant, let it work for a few minutes before trying to loosen the nut or bolt. Badly rusted fasteners may have to be chiseled off or removed with a special nut breaker, available at tool stores.

If a bolt or stud breaks off in an assembly, it can be drilled out and removed with a special tool called an E-Z out (or screw extractor). Most dealer service departments and motorcycle repair shops can perform this task, as well as others (such as the repair of threaded holes that have been stripped out).

Flat washers and lock washers, when removed from an assembly, should always be replaced exactly as removed. Replace any damaged washers with new ones. Always use a flat washer between a lock washer and any soft metal surface (such as aluminum), thin sheet metal or plastic. Special locknuts can only be used once or twice before they lose their locking ability and must be replaced.

Tightening sequences and procedures

When threaded fasteners are tightened, they are often tightened to a specific torque value (torque is basically a twisting force). Over-tightening the fastener can weaken it and cause it to break, while under-tightening can cause it to eventually come loose. Each bolt, depending on the material it's made of, the diameter of its shank and the material it is threaded into, has a specific torque value, which is noted in the Specifications. Be sure to follow the torque recommendations closely.

Fasteners laid out in a pattern (i.e. cylinder head bolts, engine case bolts, etc.) must be loosened or tightened in a sequence to avoid warping the component. Initially, the bolts/nuts should go on finger tight only. Next, they should be tightened one full turn each, in a crisscross or diagonal pattern. After each one has been tightened one full turn, return to the first one tightened and tighten them all one half turn, following the same pattern. Finally, tighten each of them one quarter turn at a time until each fastener has been tightened to the proper torque. To loosen and remove the fasteners the procedure would be reversed.

Disassembly sequence

Component disassembly should be done with care and purpose to help ensure that the parts go back together properly during reassembly. Always keep track of the sequence in which parts are removed. Take note of special characteristics or marks on parts that can be installed more than one way (such as a grooved thrust washer on a shaft). It's a good idea to lay the disassembled parts out on a clean surface in the order that they were removed. It may also be

helpful to make sketches or take instant photos of components before removal.

When removing fasteners from a component, keep track of their locations. Sometimes threading a bolt back in a part, or putting the washers and nut back on a stud, can prevent mixups later. If nuts and bolts can't be returned to their original locations, they should be kept in a compartmented box or a series of small boxes. A cupcake or muffin tin is ideal for this purpose, since each cavity can hold the bolts and nuts from a particular area (i.e. engine case bolts, valve cover bolts, engine mount bolts, etc.). A pan of this type is especially helpful when working on assemblies with very small parts (such as the carburetors and the valve train). The cavities can be marked with paint or tape to identify the contents.

Whenever wiring looms, harnesses or connectors are separated, it's a good idea to identify the two halves with numbered pieces of masking tape so they can be easily reconnected.

Gasket sealing surfaces

Throughout any motorcycle, gaskets are used to seal the mating surfaces between components and keep lubricants, fluids, vacuum or pressure contained in an assembly.

Many times these gaskets are coated with a liquid or paste type gasket sealing compound before assembly. Age, heat and pressure can sometimes cause the two parts to stick together so tightly that they are very difficult to separate. In most cases, the part can be loosened by striking it with a soft-faced hammer near the mating surfaces. A regular hammer can be used if a block of wood is placed between the hammer and the part. Do not hammer on cast parts or parts that could be easily damaged. With any particularly stubborn part, always recheck to make sure that every fastener has been removed.

Avoid using a screwdriver or bar to pry apart components, as they can easily mar the gasket sealing surfaces of the parts (which must remain smooth). If prying is absolutely necessary, use a piece of wood, but keep in mind that extra clean-up will be necessary if the wood splinters.

After the parts are separated, the old gasket must be carefully scraped off and the gasket surfaces cleaned. Stubborn gasket material can be soaked with a gasket remover (available in aerosol cans) to soften it so it can be easily scraped off. A scraper can be fashioned from a piece of copper tubing by flattening and sharpening one end. Copper is recommended because it is usually softer than the surfaces to be scraped, which reduces the chance of gouging the part. Some gaskets can be removed with a wire brush, but regardless of the method used, the mating surfaces must be left clean and smooth. If for some reason the gasket surface is gouged, then a gasket sealer thick enough to fill scratches will have to be used during reassembly of the components. For most applications, a non-drying (or semi-drying) gasket sealer is best.

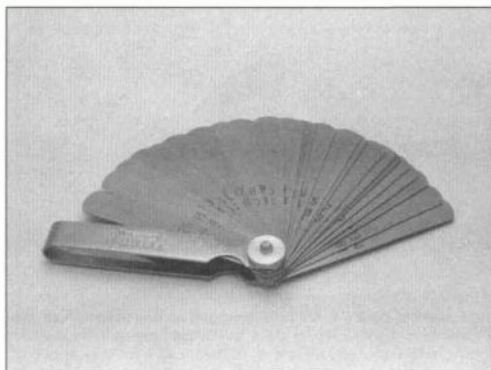
Hose removal tips

Hose removal precautions closely parallel gasket removal precautions. Avoid scratching or gouging the surface that the hose mates against or the connection may leak. Because of various chemical reactions, the rubber in hoses can bond itself to the metal spigot that the hose fits over. To remove a hose, first loosen the hose clamps that secure it to the spigot. Then, with slip joint pliers, grab the hose at the clamp and rotate it around the spigot. Work it back and forth until it is completely free, then pull it off (silicone or other lubricants will ease removal if they can be applied between the hose and the outside of the spigot). Apply the same lubricant to the inside of the hose and the outside of the spigot to simplify installation.

If a hose clamp is broken or damaged, do not reuse it. Also, do not reuse hoses that are cracked, split or torn.



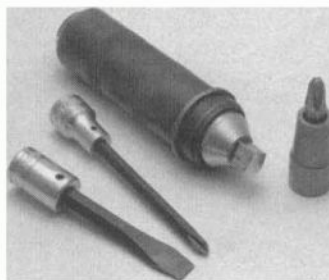
Spark plug gap adjusting tool



Feeler gauge set



Control cable pressure luber



Hand impact screwdriver and bits

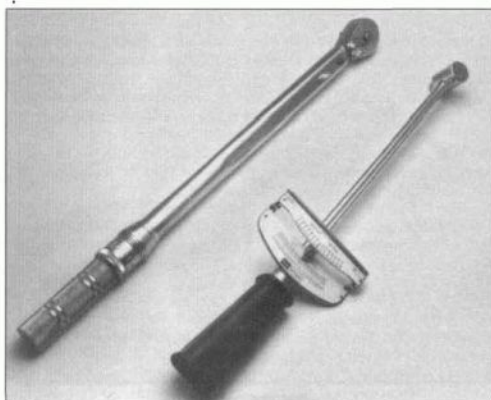
Tools

A selection of good tools is a basic requirement for anyone who plans to maintain and repair a motorcycle. For the owner who has few tools, if any, the initial investment might seem high, but when compared to the spiraling costs of routine maintenance and repair, it is a wise one.

To help the owner decide which tools are needed to perform the tasks detailed in this manual, the following tool lists are offered: Maintenance and minor repair, Repair and overhaul and Special. The newcomer to practical mechanics should start off with the Maintenance and minor repair tool kit, which is adequate for the simpler jobs. Then, as confidence and experience grow, the owner can tackle more difficult tasks, buying additional tools as they are needed. Eventually the basic kit will be built into the Repair and overhaul tool set. Over a period of time, the experienced do-it-yourselfer will assemble a tool set complete enough for most repair and overhaul procedures and will add tools from the Special category when it is felt that the expense is justified by the frequency of use.

Maintenance and minor repair tool kit

The tools in this list should be considered the minimum required for performance of routine maintenance, servicing and minor repair work. We recommend the purchase of combination wrenches (box end



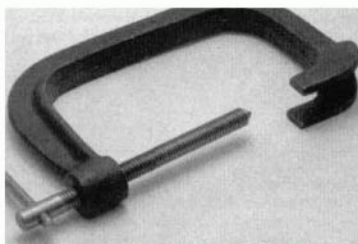
Torque wrenches (left - click type; right - beam type)



Snap-ring pliers (top - external; bottom - internal)



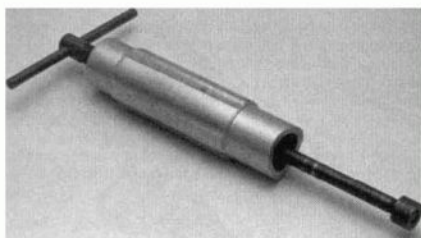
Allen wrenches (left) and Allen head sockets (right)



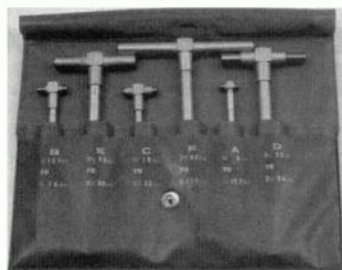
Valve spring compressor



Piston ring removal/installation tool



Piston pin puller



Telescoping gauges